Missionaries and Cannibals

Aim:

To solve the Missionaries and Cannibals problem.

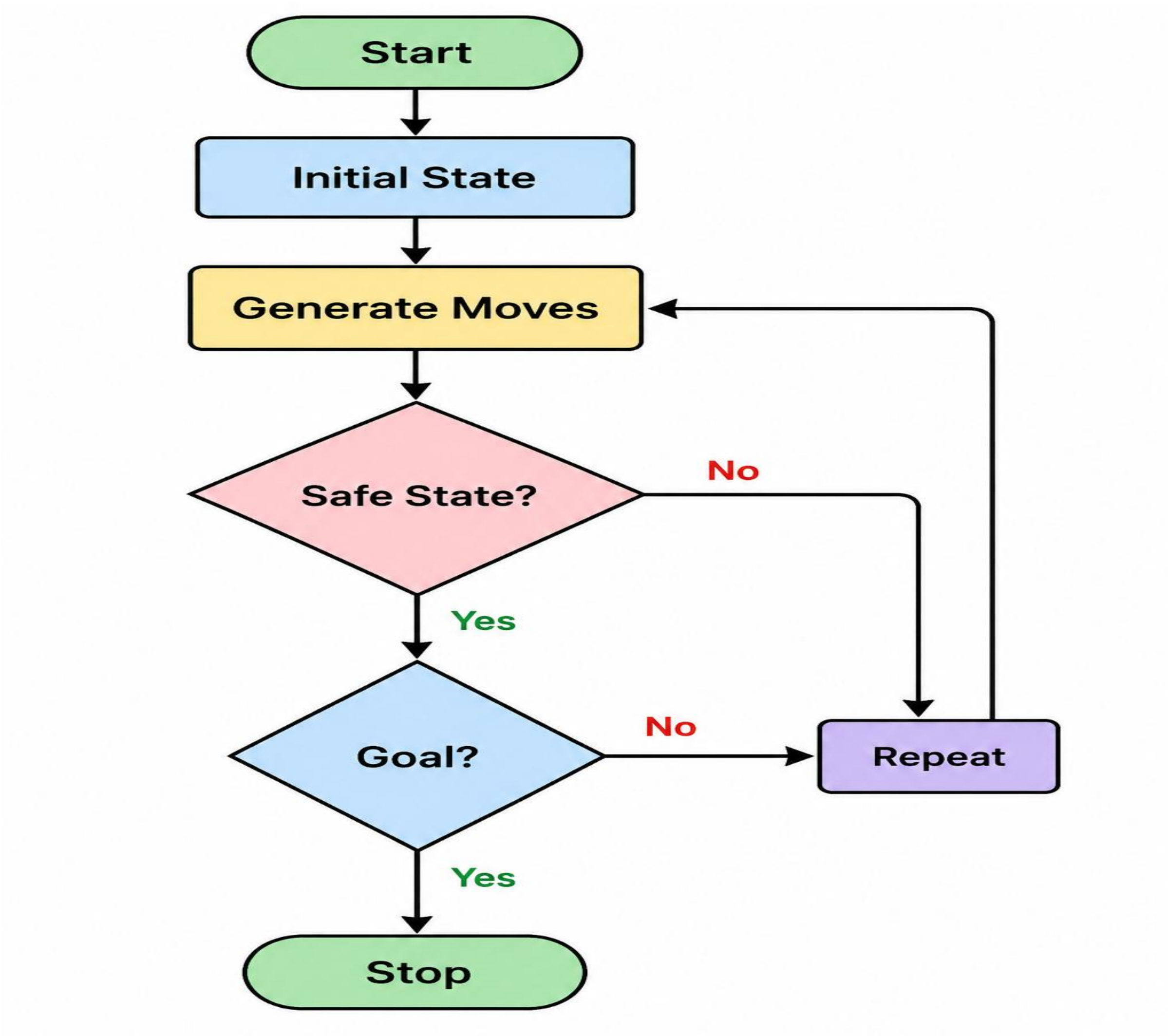
Objective:

To safely move all missionaries and cannibals across the river.

Algorithm:

1. Represent states.
2. Generate valid moves.
3. Avoid unsafe states.
4. Reach goal state.

Flowchart:



PYTHON CODE: print("Missionaries and Cannibals Solution")

```
n = int(input("Enter the number of states: "))

states = []

for i in range(n):
    state = input(f"Enter state {i+1}: ")
    states.append(state)

print("\nMissionaries and Cannibals Solution Path:")

for s in states:
    print(s)
```

OUTPUT:

```
IDLE Shell 3.13.14
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: C:/Users/Sheik Mohamed/program.py =================
Missionaries and Cannibals Solution
Enter the number of states: 4
Enter state 1: (3,3,L)
Enter state 2: (3,1,R)
Enter state 3: (3,2,L)
Enter state 4: (3,2,L)

Missionaries and Cannibals Solution Path:
(3,3,L)
(3,1,R)
(3,2,L)
(3,2,L)
>>>
```

Result

All missionaries and cannibals crossed safely.

Conclusion

The problem was solved using state-space search.